THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 04

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** may be improved or revised when new evidence appears	science is open to correction
2	**(d)** a single point	size and shape are ignored
3	**(b)** increases with temperature	as per solubility curves
4	**(d)** tiny liquid droplets in air	liquid dispersed in gas
5	**(a)** 100 u	40 + 12 + 48
6	**(c)** Al_2O_3	Al^{3+} and O^{2-} cross over
7	**(d)** 35.5 u	(35 × 0.75) + (37 × 0.25)
8	**(b)** −3 m s^{-2}	(0 − 15) ÷ 5
9	**(c)** 18 m s^{-1}	v = u + at
10	**(c)** inertia of rest	the body tries to remain at rest
11	**(d)** newton	kg m s^{-2}
12	**(c)** the magnitude or direction of the force	no energy is created
13	**(b)** vibrating objects	to and fro motion
14	**(c)** meiosis	four cells, halved chromosomes
15	**(b)** increases in weight	water enters by osmosis
16	**(d)** act as cushions and give flexibility	backbone
17	**(b)** monocots	one cotyledon
18	**(d)** uterus	womb
19	**(b)** pollination	wind, water, insects
20	**(d)** 3 × 10^8 m	distance = speed × time

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	It expresses relationships between quantities compactly and allows them to be tested carefully (2).	2
22	(a) $CaCl_2$ (b) Al_2O_3 (1+1).	1+1
23	v = gt = 10 × 2 = 20 m s^{-1} (2).	2
24	Net force = 10 − 6 = 4 N in the direction of the larger force (2).	2
25	A cell without a well-defined nucleus and membrane-bound organelles (1); bacteria (1).	1+1
26	A place where two bones meet (1); ball and socket, hinge (1).	1+1
27	Release of an egg from the ovary (1); the thick uterine lining is shed as menstruation (1).	1+1
28	Chemicals from refrigerators and aerosols that destroy ozone (1); reduced their use so the layer is recovering (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Magnet removes iron filings (1); add water, filter to remove sand (1); evaporate the filtrate to obtain salt (1).	1+1+1
30	Z = 35 − 18 = 17, chlorine (1.5); 2, 8, 7 (1.5).	1½+1½
31	A longer path means a smaller slope, hence a larger MA and less effort (3).	3
32	Quality of sound (1); overtone patterns differ because of the shape, material and construction (2).	1+2
33	Free movement in many directions requires joints such as ball and socket (1.5); shoulder and hip, knee hinge (1.5).	1½+1½
34	Local names differ and cause confusion (1); genus name capital and species name small; italics or underlined (2).	1+2
35	O (2,6) needs two electrons; each H (1) needs one (1); O shares one electron with each of two H atoms (1); H_2O has two single bonds (1)—each atom becomes stable.	1+1+1

	OR: When hydrogen is the first element no prefix is used, so H_2S is hydrogen sulfide (1.5); H_2O is water and NH_3 is ammonia — known by common names (1.5).	1½+1½
36	Evaporation from oceans, condensation, precipitation (1.5); runoff and infiltration link the hydrosphere, atmosphere, geosphere and biosphere (1.5).	1½+1½
	OR: Nitrogen fixation by bacteria and lightning (1); plants absorb and animals eat plants (1); decomposers return ammonia and denitrifying bacteria return N_2 (1).	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) NaCl solution (1). (ii) It forms free ions Na^+ and Cl^-; sugar does not form ions (2). (iii) NaCl has a high melting point; sugar has low (1). (iv) NaCl ionic; sugar covalent (1).	1+2+1+1
	OR: (i) Bond: transfer vs sharing of electrons (1); state: crystalline solids vs gases/liquids/soft solids (1); melting point: high vs low (1); conductivity: conduct when molten/dissolved vs non-conductors (1). (ii) Water molecules stabilise the ions and separate them; covalent compounds have no ions (1).	4+1
38	(i) Force opposing relative motion between surfaces in contact (1). (ii) Pull a block with a spring balance on surfaces; reading when it just moves is the friction (2). (iii) Foot pushes back, friction pushes forward (2).	1+2+2
	OR: (i) Balanced: net force zero; unbalanced: non-zero (2). (ii) 100 N towards the first team (2). (iii) First team, as net force is in its direction (1).	2+2+1
39	(i) Genetically regulated death of cells (1). (ii) Cancer cells divide without control and spread (2). (iii) To avoid excess cells, tumours and loss of function (2).	1+2+2
	OR: (i) Nucleus, organelles, size, DNA form (circular vs linear) (3). (ii) Bacteria; plant or animal cell (1). (iii) Nucleoid (1).	3+1+1
40	(i) Prevents unplanned pregnancy and infections (1). (ii) HIV/AIDS, herpes; use of condoms and avoiding contact with infected persons (2). (iii) Condoms, pills, copper-T with how each works (2).	1+2+2
	OR: (i) Ovulation and thickening of the lining and menstruation (2). (ii) The zygote implants in the uterus and pregnancy starts (2). (iii) It varies from 21 to 35 days among women (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER COPY